

Municipal Solid Waste Route Optimizer:

Use case Flow:

Use Case ID

UC-02

Use Case Name

Generate Optimized Route

Primary Actor

Sanitation Supervisor

Supporting Actor

Map Service

Main Success Scenario

1. The Sanitation Supervisor opens the Municipal Solid Waste Route Optimizer.
2. The system displays the available waste collection locations.
3. The Sanitation Supervisor selects the locations that require waste collection.
4. The Sanitation Supervisor selects the available truck for the collection route.
5. The system validates the selected locations and truck details.
6. The system sends the selected location details to the Map Service.
7. The Map Service calculates the optimized route.
8. The system receives the optimized route from the Map Service.
9. The system displays the collection locations in the optimized order.
10. The Sanitation Supervisor reviews the generated route.
11. The system saves the optimized route.
12. The use case ends successfully.

Alternate Flow: Route Generation Failed

7a. The Map Service is unable to generate an optimized route.

1. The system displays a route generation failure message.
2. The Sanitation Supervisor checks the selected collection locations.
3. The Sanitation Supervisor modifies the location selection if required.
4. The Sanitation Supervisor requests the system to generate the route again.
5. The flow returns to Step 5 of the Main Success Scenario.

